Continuous Data Assimilation for Geophysical Flow

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Research Goals

- We consider continuous data assimilation, a numerical method which incorporates known data, for the barotropic vorticity model of geophysical flow.
- We use MATLAB with the finite element C++/MATLAB toolbox FELICITY for numerical experimentation. [3]
- Guiding Question: Does incorporating only real-time vorticity data into the simulation lead to long-time stable and accurate approximate solutions even with incorrect initial conditions?

Introduction

The Barotropic Vorticity Model for Geophysical Flow, is shown in the below system of equations [2]:

$$\begin{aligned} R_0 \frac{\partial \omega}{\partial t} + R_0\, J(\psi,\omega) - \frac{\partial \psi}{\partial x} - \left(\frac{\delta_m}{L}\right)^3 \Delta\omega &= F, \quad \text{in } (0,T]\times\Omega, \\ -\Delta\psi &= \omega \quad \text{in } (0,T]\times\Omega, \\ \omega(0,\mathbf{x}) &= \omega_0 \quad \text{in } \Omega, \\ \omega(t,\mathbf{x}) &= 0 \quad \text{on } \partial\Omega, \end{aligned}$$

- ω : the vorticity function
- ψ : the streamfunction
- R_0 : the Rossby number
- δ_m : the Munk scale
- L : the length scale
- F : the forcing term
- $\mathrm{J}(\psi,\omega) = \frac{\partial\psi}{\partial x}\frac{\partial\omega}{\partial y} - \frac{\partial\omega}{\partial x}\frac{\partial\psi}{\partial y}$
- Ω : the spacial region

With Continuous Data Assimilation [1], the system becomes

$$\begin{aligned} R_0 \frac{\partial \tilde{\omega}}{\partial t} + R_0\, J(\tilde{\psi},\tilde{\omega}) - \frac{\partial \tilde{\psi}}{\partial x} - \left(\frac{\delta_m}{L}\right)^3 \Delta\tilde{\omega} + \xi &= F \quad \text{in } (0,T]\times\Omega, \\ -\Delta\tilde{\psi} &= \tilde{\omega} \quad \text{in } (0,T]\times\Omega, \\ \mu I_H(\tilde{\omega}-\omega) &= \Delta\xi \quad \text{in } (0,T]\times\Omega, \\ \tilde{\omega}(0) &= \tilde{\omega}_0 \quad \text{in } \Omega, \end{aligned}$$

- $\tilde{\omega}, \tilde{\psi}$: approximations to ω, ψ above
- I_H : a linear interpolant for CDA, where H relates to the CDA grid size
- μ : the nudging parameter

We solve the system using a finite element method for the spatial discretization and a backward Euler for the time discretization. We assume that the stream and vorticity functions are of the following form, and use these assumed solutions to determine the forcing term with which we run our numerical experiments:

$$\psi(x,t) = \exp\left(\frac{-2\pi^2(\frac{\delta_m}{L})^3 t}{R_0}\right)\sin(\pi x)\sin(2\pi y), \quad \omega(x,t) = 5\pi^2\psi.$$

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Evolution of Solutions with CDA

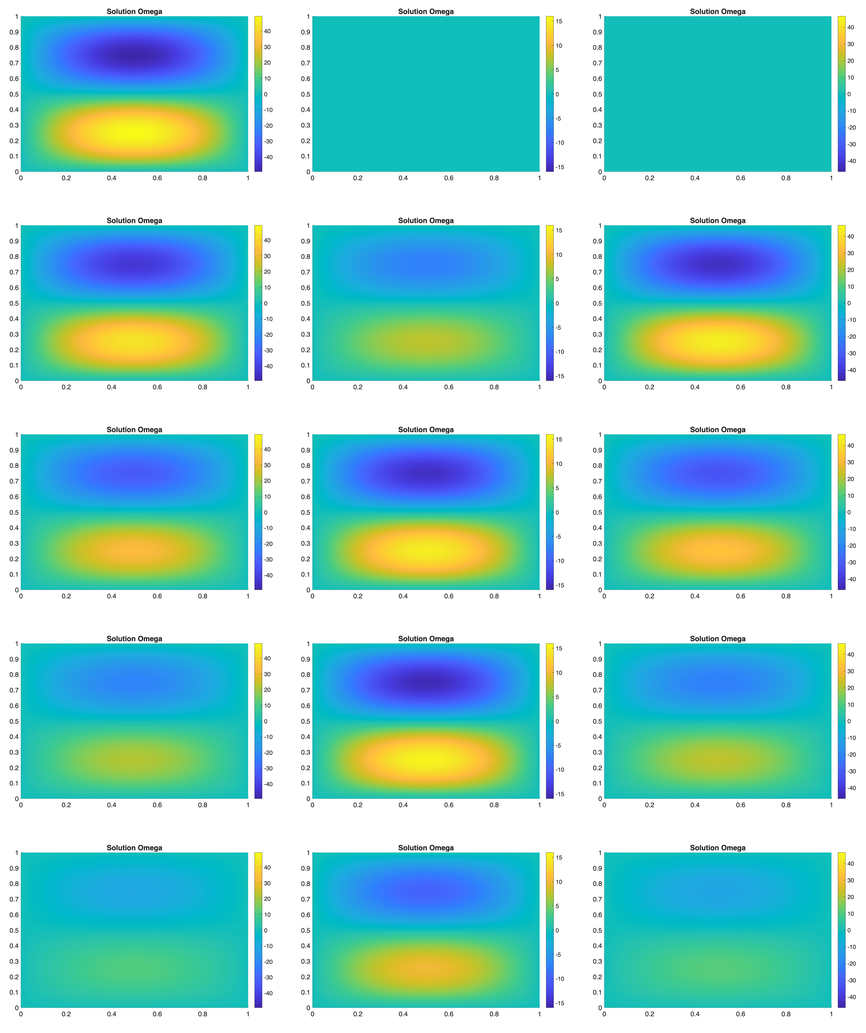


Figure: The vorticity function at time steps 0, 20, 60, 120, and 220 for (left) the true solution, (center) the numerical simulation with incorrect initial conditions and no data, and (right) the numerical simulation with incorrect initial conditions and some data, $H = 1/16, \mu = 1000$.

References

[1] A. Azouani, E. Olson, and E.S. Titi. Continuous data assimilation using general interpolant observables. *Journal of Nonlinear Science*, 24:277–304, 2014.

[2] D.D. Holm and B.T. Nadiga. Modeling mesoscale turbulence in the barotropic double-gyre circulation. *Journal of Physical Oceanography*, 33(11):2355–2365, 2003.

[3] S.W. Walker FELICITY: A Matlab/C++ Toolbox for Developing Finite Element Methods and Simulation Modeling. *SIAM Journal on Scientific Computing*, 40 (2):C234-C257, 2018.

FEM Mesh and Data Grid

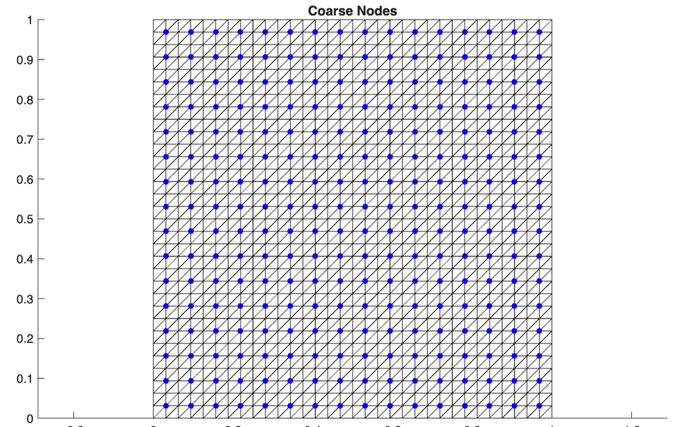
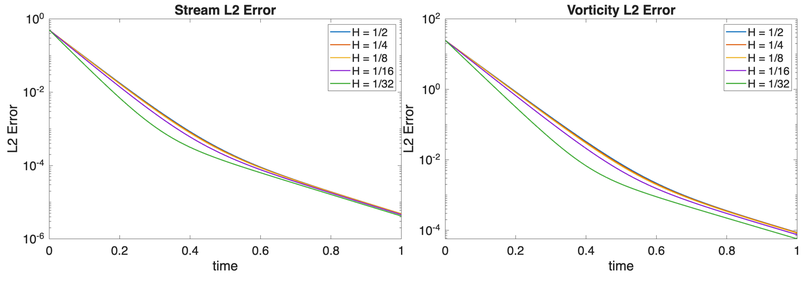


Figure: Black: FEM Mesh $h = 1/64$, Blue: Data Grid $h = 1/16$

Long-Time Accuracy

L2 Error for Various Grid Size



L2 Error for Various Nudging Parameters

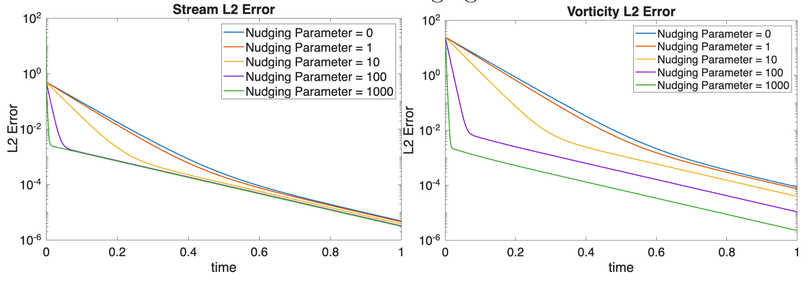


Figure: fine mesh size $= 1/64$, time step size $= 0.001$, $R_0 = 1$, $(\delta_M/L) = 0.7$, $\mu = 1$ (top), data grid size $= 1/16$ (bottom)

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